



The role of obesity-risk perception and gender differences in colorectal cancer screening among eligible adults in the United States

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ABSTRACT

Early detection of colorectal cancer (CRC) is crucial in reducing the associated morbidity and mortality. Our study aims to assess the association between obesity-cancer risk perception and the uptake of CRC screening among eligible adults in the United States. Additionally, we explored the variation of association stratified by gender. We analyzed data from the Health Information National Trends Survey (HINTS) 2018–2020. We performed descriptive statistics, bivariable, and multivariable logistic regression analysis. Of the 6435 overweight and obese US adults ≥ 45 years, 69 % ($n = 4970$) performed CRC screening. Approximately 85.4 % ($n = 3243$) of participants who believed that obesity increased the risk of cancer were screened. Perceived risk of obesity-related cancer was associated with 35 % significantly higher odds of screening (cOR:1.35,95 %CI:1.03–1.75). When compared to participants with no insurance, participants with health insurance had significantly higher odds of undergoing CRC screening (aOR:4.41,95 %CI:1.87–10.40). Non-Hispanic Black individuals had significantly higher odds of CRC screening compared to non-Hispanic White individuals (aOR: 2.17, 95 % CI: 1.04–4.51). For every one-year increase in age, there is an 11 % increase in screening (aOR:1.11,95 % CI:1.07–1.16). There was no significant difference in CRC screening rates between males and females. Non-Hispanic Black females had significantly higher adjusted odds of CRC screening (aOR:3.17, 95 %CI:1.23–8.19) compared to non-Hispanic White females. Female college graduates had significantly higher odds of screening when compared to high school graduates (aOR:2.86,95 %CI:1.07–7.61). When compared to males who are non-smokers, former smokers had significantly higher odds of undergoing CRC screening (aOR:2.97,95 % CI:1.19–7.45). Male homosexuals/gay/lesbian when compared to heterosexual/straight males had significantly higher odds to be screened for CRC (aOR: 10.17,95 % CI:1.13–91.10). This study provided essential insights that help bridge gaps in cancer disparities, revealing that race/ethnicity and insurance status significantly impact CRC screening uptake. Health promotion programs should implement tailored interventions that emphasize perceived susceptibility to obesity and consider gender-specific factors to increase CRC screening rates.

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Introduction

Colorectal cancer (CRC) ranks as the third most commonly diagnosed cancer in the United States (US) [1]. Approximately 70 % of CRCs are found in the colon, while 30 % occur in the rectum [2]. According to the National Cancer Institute (NCI), the estimated number of new cases is 152,810, and the estimated number of deaths is 53,010 in 2024 [3]. The rate of new cases of colorectal cancer was 36.5 per 100,000 men and women per year and the death rate was 12.9 per 100,000 men and women per year [3]. The age-adjusted incidence and mortality rates for colorectal cancer are higher among men than women [3]. According to the Centers for Disease Control and Prevention (CDC), men have an incidence and mortality rate of 40.7 per 100,000 and 15.0 per 100,000, respectively [4], while women have an incidence rate of 32.0 per 100,000 and a mortality rate of 10.6 per 100,000 [4].

According to the American Cancer Society (ACS), the risk factors for CRC are broadly divided into modifiable and non-modifiable risks where modifiable risk factors include obesity, diabetes mellitus, smoking, alcohol use, and diets high in red and processed meats and non-modifiable risk factors include age, sex, history of cholecystectomy, family history of colorectal cancer or adenomatous polyps, radiation to the abdomen or pelvic area, personal history of inflammatory bowel diseases, and colorectal polyps [5]. Early detection of CRC through screening is essential for improving CRC outcomes [6], as subsequent removal of polyps can help prevent the disease [7].

In 2018, ACS recommended qualified screening for adults starting at age 45 years [8]. While research suggests that CRC screening rates in the U.S. have increased following the implementation of the Affordable Care Act (ACA) [9], disparities persist with screening rates remaining low, particularly among the uninsured, individuals in medically underserved communities, and minority populations [10]. Additionally, studies have shown that factors associated with low screening include having a low level of education, living in poverty, lacking health insurance, and English as a second language [11–14].

Although racial disparities in CRC screening continue to persist, the specific direction of these disparities remains unclear due to inconsistencies in the literature. Some studies suggest that non-Hispanic Black individuals have higher odds of CRC screening [15,16], while others report higher screening rates among non-Hispanic White individuals [17,18]. Additionally, some studies have found no significant difference in screening rates between non-Hispanic Whites and Blacks [19]. These mixed findings highlight the need for further research to better understand racial disparities in CRC screening.

Gender differences in colorectal cancer (CRC) screening have been understudied, and existing findings are inconsistent. Studies suggest women participate in general cancer screening more than men [20]. This is not surprising because men tend to use fewer medical services than women [21]. Davis et al. attribute gender differences in screening to women's more frequent health visits, heightened media focus on female health, and men's doubts about screening benefits for CRC cancer [22]. Several studies suggest that CRC screening rates are higher among males than females [23–25]. However, some studies report no significant gender differences in CRC screening rates [16]. Additionally, race-gender-based disparities have been observed, with Hispanic men showing the lowest CRC screening rates, followed by Asian, American Indian/Alaska Native, Black, and White men. Among women, non-Hispanic Black females are more likely to undergo CRC screening compared to White females [16].

Psychosocial factors at the patient level have been shown to influence participation in colorectal cancer (CRC) screening. Reported barriers to CRC screening include fear of test results, the burden of bowel preparation, lack of information, time constraints, low self-worth, sensitivities related to "para-sexual" concerns, fatalistic attitudes, negative prior experiences with testing, and distrust of the financial motives behind screening recommendations [26]. Other psychosocial influences, such as social pressure, cultural norms, and health beliefs, including

perceived susceptibility, can act as obstacles impacting the initiation of and adherence to CRC screening behaviors [27].

The Health Belief Model (HBM) frequently identifies factors influencing preventive health behaviors, including vaccination, safe sexual practices, and cancer screenings [28,29]. A review of multiple studies demonstrated that the HBM effectively explains the determinants of CRC screening behaviors [30]. Some studies that focused on predictors of colonoscopy compliance have shown that perceived susceptibility to colorectal polyps, perceived severity, perceived barrier that the screening test had not been recommended by a healthcare provider (HCP), and perceived benefits of CRC screening are associated with colonoscopy and flexible sigmoidoscopy screening [31–34]. Overweight and obesity are established risk factors for CRC [35]. Research indicates that a high body mass index (BMI) in early life is linked to an increased risk of CRC in both young people and adults [36,37]. Okabayashi et al. found a dose-response relationship between BMI and the prevalence of colorectal adenomas, with higher BMI categories associated with greater risk [38]. Another study has shown that sustained high BMI, especially long before diagnosis, is a strong CRC risk factor [39]. Additionally, being overweight and obese is associated with a higher likelihood of receiving CRC screening [15]. However, a study reported no association between being overweight or obese and the perception of increased risk for colorectal cancer [40].

While several studies have linked obesity to an increased risk of CRC [35–39], there is still a paucity of research that has focused on how perceptions of obesity-associated risk of CRC influence screening behaviors among obese and overweight. Given that few studies have identified determinants of screening uptake separately for men and women, and the recommended age range for screening has also changed, we hypothesize that trends in the variation may have changed. Therefore, our study aims to fill this gap by assessing the association between obesity-cancer risk perception and uptake of CRC screening tests, along with the impact of gender on CRC screening participation.

Methods

Data source, study design, and population

The Health Information National Trends Survey (HINTS) is a nationally representative cross-sectional survey of adults aged 18 years and above in the civilian, noninstitutionalized population of the US. HINTS collects information on access to and usage of health-related information and health-related behaviors, such as perceptions and behaviors about cancer screening. The HINTS survey uses a complex sampling design to represent the adult population accurately. Details of the design, conduct, and analysis of the HINTS were published previously [41]. Using consistent weighting algorithms, we combined data from HINTS 5, cycles 2–4 (2018–2020), resulting in a total sample of 12,807 respondents. The study focused on 6435 participants aged 45 and older who were classified as overweight and obese, following the World Health Organization's Body Mass Index (BMI) Classification [42]. Since the study had no direct patient contact and all subjects were de-identified, it did not require Institutional Review Board (IRB) approval. The study is presented per the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines [43].

Measures

Outcome variable: screened for colorectal cancer

Participants were asked, "There are a few tests to check for colon cancer. These tests include colonoscopy, sigmoidoscopy, and stool blood test. Have you ever had one of these tests to check for colon cancer?" The outcome variable "Screened for CRC," was categorized as "Yes" or "No."

Main exposure

The main exposure was obesity-cancer risk perception. To assess

obesity-cancer risk perception status, respondents were asked: “How much do you think any of the following can influence whether or not the person will develop cancer: being overweight or obese?” Responses included “A lot,” “A little,” “Not at all,” or “Don’t know.” The responses were categorized into ‘A lot/a little’, and ‘not at all.’ Those who responded with “Don’t know” were excluded from the study.

Covariates

Based on previous literature, the sociodemographic characteristics, disease condition, lifestyle, and behavioral characteristics included in the study were age (continuous, in years), gender (male or female), health insurance (yes or no), education (high school or less or some college or college graduate and more), sexual orientation (heterosexual or straight or homosexual/gay/lesbian or bisexual), race/ethnicity (non-Hispanic White or non-Hispanic Black or Hispanic or non-Hispanic Asian/non-Hispanic other), household income (< \$50,000 or ≥ \$50,000), family history of cancer (yes or no), diabetes (yes or no), number of chronic disease conditions (absence or presence), smoking status (never smoked or current smoker or former smoker) and physical activity (< 150 Min per Week or ≥ 150 Min per Week)

Statistical analysis

We used weighted frequencies and percentages to present participants’ sociodemographic characteristics, disease conditions, and lifestyle and behavioral traits by CRC screening status in the overall population and further stratified by gender. We used Pearson’s Chi-square test to assess the relationship between colon cancer screening and independent variables. Since the outcome variable was binary, we applied logistic regression. We conducted univariable and multivariable logistic regression analyses to evaluate the association between colorectal cancer screening and covariates in the overall population and stratified by gender. We computed odds ratios (ORs) with corresponding 95 % confidence intervals (CIs) to measure the strength of the association between the explanatory variables and the outcome variable. All missing data were coded as missing and were included in the regression analysis. All the covariates were included in the study. We conducted statistical analyses using SAS software (version 9.4), utilizing the PROC SURVEYFREQ procedure with the Taylor series linearization method to estimate standard errors, given the complexity of the survey data. Statistical significance was two-sided and set at p-value < 0.05.

results

Sociodemographic characteristics of study population

In Table 1, of the 6435 participants included in the analysis, 4970 (69.0 %) had CRC screening, while 1329 (31.0 %) had not undergone CRC screening. The majority (85.4 %) of participants who perceived a risk of obesity-related cancer underwent screening. Among those who were screened, more than half were non-Hispanic White (71.9 %), had a family history of cancer (79.6 %), and never smoked (58.0 %). A majority had insurance (97.8 %) and were Heterosexual or straight (96.8 %).

Among males who have been screened, the majority had a perceived risk of obesity-related cancer (86.5 %), were Heterosexual/straight (95.8 %), and had insurance (97.9 %). More than half of the screened males were non-Hispanic Whites (75.5 %), had a family history (78.5 %), and never smoked (52.5 %). Among female respondents who have been screened for CRC, 84.3 % had a perceived risk of obesity-related cancer, 97.9 % were heterosexual or straight, 97.7 % were insured, and 80.9 % had a family history of cancer (Table 2).

Association between factors and CRC screening in the overall population

In Table 3, factors such as the perceived risk of obesity-related

Table 1
Descriptive statistics in the overall study population stratified by CRC screening status - HINTS 2018–2020.

Overall sample size (N = 6435)				
Characteristic	Overall sample (N = 6435) n(wt%)	Colon Cancer Screening (n = 6299)		P-value
		Screened n(wt%)	Unscreened n(wt%)	
Screening status				
Screened	4970 (69.0 %)			
Unscreened	1329 (31.0 %)			
Perceived Risk of Obesity-Related Cancer				0.03
Yes	4104 (84.3)	3243(85.4)	796 (81.4)	
No	644 (15.7)	470 (14.6)	161 (18.6)	
Age, Mean (SE)	59.95(0.22)	62.79(0.25)	53.57(0.29)	
Gender				0.2180
Male	2948(52.3)	2313(51.7)	605 (54.5)	
Female	3430(47.7)	2640(48.3)	719(45.5)	
Sexual orientation				0.0628
Heterosexual or straight	5834(97.2)	4520(96.8)	1206(98.1)	
Homosexual or Gay or Lesbian	122 (2.2)	95(2.6)	23(1.3)	
Bisexual	51(0.6)	37(0.6)	13(0.6)	
Insurance				<.0001
Yes	6107(94.5)	4811(97.8)	1171(86.8)	
No	233(5.5)	94(2.2)	135(13.2)	
Education				0.0002
High school or less	2344(45.4)	1731(43.0)	550(50.7)	
Some college	1556(30.7)	1226(30.9)	307(30.4)	
College graduates and more	2454(23.9)	1963(26.1)	455(18.9)	
Race/Ethnicity				<.0001
Non-Hispanic White	3654 (67.6)	2904(71.9)	692(58.5)	
Non-Hispanic Black	967 (12.8)	759(12.7)	178(11.9)	
Hispanic	836 (14.1)	574(10.4)	250(22.3)	
Non-Hispanic Asian/ Non-Hispanic Other	325 (5.6)	226(5.0)	98(7.3)	
Household Income				0.0597
< \$50,000	2769 (42.9)	2077(41.2)	628(46.2)	
≥ \$50,000	2989 (57.1)	2352(58.8)	595(53.8)	
Family History of Cancer				0.1138
Yes	1385 (78.0)	1111 (79.6)	233 (72.5)	
No	345 (22.1)	252 (20.4)	86 (27.6)	
Diabetes				0.0203
Yes	1987 (29.1)	1626(31.0)	315(24.9)	
No	4371 (70.9)	3290(69.0)	997(75.2)	
Number of Chronic Conditions				<.0001
Absence	1789 (33.5)	1212(29.0)	543(43.8)	
Presence	4493 (66.5)	3647(71.0)	754(56.2)	
Physical Activity				0.7244
< 150 Minutes per Week	4477(69.0)	3445(69.1)	929(68.3)	
≥ 150 Minutes per Week	1958(31.0)	1525(30.9)	400(31.7)	
Smoking Status				<.0001
Never Smoked	3731(57.8)	2874 (58.0)	785 (57.3)	
Current smoker	724 (13.6)	471 (10.7)	232 (19.8)	
Former smoker	1897(28.7)	1566 (31.3)	300 (22.9)	

p-value:

*p < 0.05,

** < 0.01,

*** < 0.001

wt%= Weighted percentage

cancer, age, insurance, college graduate, education, number of chronic diseases, and being a former smoker were significantly associated with CRC screening. We found that those with a perceived risk of obesity-related cancer had 1.35 (95 % CI:1.03–1.75) times higher unadjusted odds of colon cancer screening compared to those who had no perceived risk. College graduates had 63 % significantly higher odds of colon cancer screening (cOR: 1.63;95 %CI:1.31–2.02) when compared to

Table 2

Descriptive statistics of CRC Screening status in the overall population stratified by gender - HINTS 2018–2020.

Characteristics	Male (n = 2948)		P-value	Female (n = 3430)		P-value
	Screened (n = 2313) n(wt%)	Unscreened (n = 605) n(wt%)		Screened (n = 2640) n(wt%)	Unscreened (n = 719) n(wt%)	
Perceived Risk of Obesity-Related Cancer			0.0684			0.2374
Yes	1563(86.5)	377(81.5)		1670(84.3)	417(81.3)	
No	210(13.5)	66(18.5)		259(15.7)	94(18.7)	
Age, Mean (SE)	62.16(0.35)	53.16(0.43)		63.43(0.35)	54.03(0.46)	
Sexual orientation			0.0668			0.7103
Heterosexual or straight	2103 (95.8)	558 (98.1)		2404(97.9)	644(98.0)	
Homosexual or Gay or Lesbian	64 (3.4874)	9 (1.2930)		31(1.7)	14(1.3)	
Bisexual	21 (0.7152)	7 (0.5761)		16(0.4)	6(0.7)	
Insurance			<.0001			<.0001
Yes	2242(97.9)	527 (85.4)		2554(97.7)	641(88.6)	
No	37(2.1)	68(14.6)		56(2.3)	66(11.4)	
Education			<.0001			0.7906
High school or less	730 (41.6)	252 (54.0)		995 (44.5)	294 (46.5)	
Some college	539 (29.2)	147 (29.2)		684 (32.8)	160 (32.0)	
College graduates and more	1020 (29.3)	200 (16.8)		937 (22.7)	254 (21.5)	
Race/Ethnicity			<.0001			0.0002
Non-Hispanic White	1468 (75.5)	338 (58.4)		1431 (68.0)	353 (58.9)	
Non-Hispanic Black	251 (9.4)	59 (10.1)		507 (16.4)	118 (14.2)	
Hispanic	277 (10.5)	116 (24.2)		296 (10.5)	133 (20.2)	
Non-Hispanic Asian/Non-Hispanic Other	114 (4.9)	44 (7.4)		111 (5.1)	52 (6.8)	
Household Income			0.0055			0.8213
< \$50,000	794 (34.3)	269 (44.7)		1272(48.9)	357(48.1)	
≥ \$50,000	1310 (65.7)	299 (55.3)		1039(51.1)	295(51.9)	
Family History of Cancer			0.1244			0.4441
Yes	492 (78.5)	110 (68.9)		616(80.9)	123(76.6)	
No	132 (21.5)	46 (31.1)		120(19.1)	40(23.4)	
Diabetes			0.1766			0.0182
Yes	764(30.5)	138(25.2)		854(31.5)	176(24.5)	
No	1528(69.5)	457(74.8)		1754(68.5)	536(75.5)	
Number of Chronic Conditions			0.0001			<.0001
Absence	556 (30.0)	254 (42.1)		653 (28.0)	288(45.7)	
Presence	1712 (70.0)	336 (57.9)		1923 (72.0)	414 (54.3)	
Physical Activity			0.6437			0.2973
< 150 Minutes per Week	1434(63.4)	401(65.2)		1995 (75.2)	525 (72.2)	
≥ 150 Minutes per Week	879(36.6)	204(34.8)		645 (24.8)	194 (27.8)	
Smoking Status			<.0001			0.0521
Never Smoked	1216 (52.5)	329 (53.2)		1649(63.8)	453(62.4)	
Current smoker	223 (11.4)	119 (24.3)		247(10.0)	112(14.5)	
Former smoker	850 (36.1)	151(22.6)		710(26.2)	148(23.1)	

p-value:

*p < 0.05,

* * < 0.01,

* * * < 0.001

wt%= Weighted percentage

participants with some college degree. Compared to never-smokers, former smokers (cOR:1.36; 95 % CI:1.07–1.72) had significantly higher odds of being screened. Insured participants were more likely to be screened for CRC than uninsured (cOR: 6.70; 95 % CI: 4.48–10.03).

The multivariable analysis showed that compared to participants without a perceived risk of obesity-related cancer, those with a perceived risk showed no significant difference in the odds of undergoing colon cancer screening (aOR:1.49; 95 % CI:0.64–3.44). For every one-year increase in age, there is an 11 percent increase (aOR:1.11; 95 % CI: 1.07–1.16) in the odds of screening. Insured compared to uninsured participants (aOR:4.41;95 %CI:1.87–10.40) and non-Hispanic Blacks compared to non-Hispanic Whites (aOR: 2.17; 95 % CI:1.04–4.51) had significantly higher odds of being screened (Table 3).

Association between factors and CRC screening by gender

Following stratification by gender, among males, the result of the bivariate analysis showed that age, insurance, college graduate status, household income, presence of chronic disease, and former smoking were significant predictors of CRC screening. Insured males had 7.85 (95 % CI:4.30–14.35) significantly higher odds of CRC screening.

College graduates had 27 % significantly higher odds of colon cancer screening (cOR: 2.27; 95 % CI:1.65–3.13). Men with household income ≥ \$50,000 (cOR:1.55; 95 % CI:1.14–2.11) and never-smokers (cOR: 1.62; 95 % CI: 1.13–2.32) had significantly higher odds of being screened.

Among males, the result of the multivariable regression analysis showed that compared to heterosexual or straight, homosexual/gay/lesbian had 10.17 (aOR:95 % CI:1.13–91.10) significantly higher odds of being screened. Individuals who are former smokers (aOR: 2.97; 95 % CI:1.19–7.45) had significantly higher odds of being screened. Participants with a perceived risk showed no significant difference in the odds of screening (aOR:1.19; 95 % CI: 0.38–3.76). (Table 4)

Insured females had 5.53 (95 % CI:3.28–9.30) significantly higher odds of colon cancer screening compared to uninsured females. Compared to participants with no chronic disease, those with chronic disease had 2.17 (95 % CI: 1.67–2.81) times significantly higher odds of screening. The result of the multivariable regression analysis showed insured females had 5.98 (95 % CI:1.25–28.57) significantly higher odds of colon cancer screening compared to uninsured. Participants with some college degree had 86 % significantly higher odds of colon cancer screening (aOR: 2.86; 95 % CI:1.07–7.61). Compared to female non-

Table 3
Association between participant characteristics and screening status - HINTS 2018–2020.

Characteristics	Crude OR (95 %CI) (Screened vs. unscreened)	Adjusted OR (95 % CI) (Screened vs. unscreened)
Perceived Risk of Obesity-Related Cancer		
No	1.0	1.0
Yes	1.35 (1.03–1.75) * **	1.49 (0.64–3.44)
Age	1.12 (1.10–1.13) * **	1.11 (1.07–1.16) * **
Gender at Birth		
Female	1.0	1.0
Male	0.89 (0.74–1.07)	0.88 (0.53–1.43)
Sexual orientation		
Heterosexual or straight	1.0	1.0
Homosexual or Gay or Lesbian	2.023(1.06–3.87)	2.92 (0.68–12.61)
Bisexual	0.971(0.40–2.35)	0.60 (0.16–2.21)
Insurance		
No	1.0	1.0
Yes	6.70 (4.48–10.03) * **	4.41 (1.87–10.40) * **
Education		
High school or less	1.0	1.0
Some college	1.20 (0.95–1.51)	1.58 (0.88–2.84)
College graduates and more	1.63(1.31–2.02) * **	1.52 (0.87–2.67)
Household Income		
< \$50,000	1.0	1.0
≥ \$50,000	1.23 (0.99–1.52)	1.31 (0.72–2.39)
Race/Ethnicity		
Non-Hispanic White	1.0	1.0
Non-Hispanic Black	0.87(0.63–1.19)	2.17 (1.04–4.51) * **
Hispanic	0.38 (0.29–0.49) * **	0.92 (0.43–1.99)
Non-Hispanic Asian/Non-Hispanic Other	0.56 (0.36–0.87) * **	1.34 (0.47–3.86)
Family History of Cancer		
No	1.0	1.0
Yes	1.49 (0.91–2.44)	1.35 (0.62–2.93)
Diabetes		
No	1.0	1.0
Yes	0.74 (0.57–0.96) * **	0.97 (0.46–2.03)
Number of Chronic Conditions		
Absence	1.0	1.0
Presence	1.91 (1.58–2.30) * **	1.36 (0.82–2.24)
Physical Activity		
< 150 Minutes per Week	1.0	1.0
≥ 150 Minutes per Week	0.96 (0.78–1.18)	1.31 (0.75–2.29)
Smoking Status		
Never Smoked	1.0	1.0
Current smoker	0.54 (0.37–0.74)	1.06 (0.47–2.42)
Former smoker	1.36 (1.07–1.72) * **	1.28 (0.75–2.18)

p-value:

*p < 0.05,

* * < 0.01,

* ** < 0.001

OR = Odds ratio

CI = Confidence Interval

Hispanic Whites, non-Hispanic Blacks had 3.17 (95 % CI:1.23–8.19) significantly higher odds of being screened. (Table 4)

Trend in screening rates stratified by gender

In the trend analysis, all the p-values were > 0.05, and there were no significant differences in colorectal cancer screening rates between males and females over the three-year study period. This finding suggests that both genders are equally likely to undergo colorectal cancer

screening within the observed time frame. The absence of a significant gender disparity may indicate a positive trend toward behavioral changes and increased awareness regarding the importance of colorectal cancer screening across both sexes in a nationally representative sample (Fig. 1).

Discussion

Our findings indicate that 69.0 % of the study participants underwent CRC screening, which is below the Healthy People 2030 national target and goal of 72.8 % [44]. Overall, age, insurance, and race were significantly associated with screening uptake in the entire population. Some factors were common for both genders; others were unique to each gender. Our findings highlight the critical role of perceived susceptibility in preventive care, demonstrating a positive association between perceiving obesity as a cancer risk factor and CRC screening. This supports prior studies emphasizing perceived risk as a crucial behavioral construct in understanding screening participation [31,35,45,46]. To maximize behavior and increase prevention practices, public health education programs should emphasize the perceived susceptibility of obesity to developing CRC.

The role of overweight and obesity as risk factors for CRC has long been established [35–39]. Our study revealed that participants who perceived obesity as a risk factor for cancer were more likely to undergo CRC screening in the unadjusted model. Furthermore, our results are consistent with studies that support theoretical models where perceived susceptibility is viewed as a more indirect predictor of intention and behavior [46] and identify perceived CRC risk as a key factor influencing CRC screening [40]. Other related factors may explain this association. Key parts of the HBM, like perceived benefits and severity, may also influence screening uptake [32,47,48]. This underscores the importance of risk perception in CRC screening decisions and in shaping HBM-based educational interventions.

After considering confounding factors, the association between perceiving obesity as a cancer risk factor and CRC screening was not significant in the adjusted model. This suggests that socioeconomic status, healthcare access, and demographic differences may play a greater role in CRC screening behavior than obesity-cancer risk perception alone. Furthermore, our results indicate that insurance is a key factor in CRC screening uptake, with a lack of insurance associated with lower odds of screening. This finding contradicts several studies suggesting that while health insurance is necessary for access, it is insufficient for enhancing screening participation [9,49]. The direction of these disparities aligns with previous research linking a lack of health insurance to significant disparities in the utilization of preventive care services [18,50]. Additionally, the passage of the Patient Protection and Affordable Care Act (ACA) of 2010, which reduced the cost of care and improved access to screening services, may help explain the positive association between insurance and CRC screening observed in our study [9,51].

Our analysis reveals significant racial disparities in screening, with non-Hispanic Whites having lower odds of screening compared to non-Hispanic Blacks, which is consistent with findings from other studies [15,16,52]. This contrasts with research indicating that non-Hispanic Blacks have historically had lower CRC screening rates than non-Hispanic Whites and other racial minorities [17,18]. However, this gap has narrowed in recent years [53], likely due to improved insurance coverage [51]. This may explain the higher screening odds for non-Hispanic Blacks in our study. Similarly, May et al. observed disparities in screening rates between non-Hispanic Whites and other racial/ethnic groups [9]. These racial disparities highlight ongoing challenges to health equity in CRC screening. National-level, and government-funded programs like the Colorectal Cancer Control Program (CRCCP) aim to increase screening among low-income groups [54], which may help explain the higher odds of screening among non-Hispanic Blacks in our study.

Table 4

Association between participant characteristics and screening status stratified by gender - HINTS 2018–2020.

Characteristics	Male		Female	
	Crude OR (95 %CI) (Screened Vs Unscreened)	Adjusted OR (95 %CI) (Screened Vs Unscreened)	Crude OR (95 %CI) (Screened Vs Unscreened)	Adjusted OR (95 %CI) (Screened Vs Unscreened)
Perceived Risk of Obesity-Related Cancer				
No	1.0	1.0		1.0
Yes	1.46 (0.97–2.18)	1.19(0.38–3.76)	1.24 (0.87–1.79)	1.60 (0.47–5.40)
Age	1.12 (1.10–1.15) * **	1.14 (1.09–1.20) * **	1.11(1.09–1.13) * **	1.10 (1.04–1.17) * *
Sexual orientation				
Heterosexual or straight	1.0	1.0	1.0	1.0
Homosexual or Gay or Lesbian	2.76 (0.99–7.74)	10.17(1.13–91.10) * **	1.24 (0.55–2.81)	1.05 (0.08–13.93)
Bisexual	1.27(0.37–4.33)	0.47(0.07–3.27)	0.68(0.20–2.30)	0.65(0.07–6.52)
Insurance				
No	1.0	1.0	1.0	1.0
Yes	7.85(4.30–14.35) * **	4.35(1.03–18.38) * *	5.53(3.28–9.30) * **	5.98(1.25–28.57) * **
Education				
High school or less	1.0	1.0	1.0	1.0
Some college	1.30 (0.92–1.84)	1.06 (0.44–2.54)	1.07 (0.78–1.47)	2.86 (1.07–7.61) * **
College graduate and more	2.27 (1.65–3.13) * **	2.16 (0.90–5.16)	1.11 (0.82–1.49)	1.20 (0.52–2.78)
Household Income				
< \$50,000	1.0	1.0	1.0	1.0
≥ \$50,000	1.55 (1.14–2.11) * **	1.68 (0.64–4.42)	0.97 (0.72–1.30)	0.85(0.39–1.85)
Race/Ethnicity				
Non-Hispanic White	1.0		1.0	1.0
Non-Hispanic Black	0.72(0.40–1.28)	1.09 (0.34–3.56)	1.00 (0.71–1.42)	3.17 (1.23–8.19) * **
Hispanic	0.33(0.23–0.47)	1.53 (0.58–4.04)	0.45 (0.32–0.64)	0.52 (0.16–1.74)
Non-Hispanic Asian/Non-Hispanic Other	0.51(0.28–0.93)	0.54 (0.14–2.06)	0.65 (0.33–1.26)	2.19 (0.37–12.89)
Family History of Cancer				
No	1.0	1.0	1.0	1.0
Yes	1.64 (0.86–3.16)	1.61 (0.63–4.14)	1.29 (0.67–2.47)	1.52 (0.53–4.42)
Diabetes				
No	1.0	1.0	1.0	1.0
Yes	0.77 (0.53–1.13)	0.94(0.37–2.42)	0.71 (0.53–0.94)	1.10 (0.43–2.78)
Number of Chronic Conditions				
Absence	1.0	1.0	1.0	1.0
Presence	1.70(1.30–2.22) * **	1.03 (0.46–2.32)	2.17 (1.67–2.81) * **	1.89(0.91–3.92)
Physical Activity				
< 150 Minutes per Week	1.0	1.0	1.0	1.0
≥ 150 Minutes per Week	1.08 (0.77–1.51)	1.43 (0.63–3.26)	0.86 (0.64–1.15)	1.67(0.70–3.98)
Smoking Status				
Never Smoked	1.0	1.0	1.0	1.0
Current smoker	0.48 (0.29–0.77)	1.49(0.46–4.79)	0.67 (0.47–0.96)	0.62 (0.19–1.96)
Former smoker	1.62 (1.13–2.32) * **	2.97(1.19–7.45) * **	1.11 (0.82–1.50)	0.47 (0.19–1.20)

p-value:

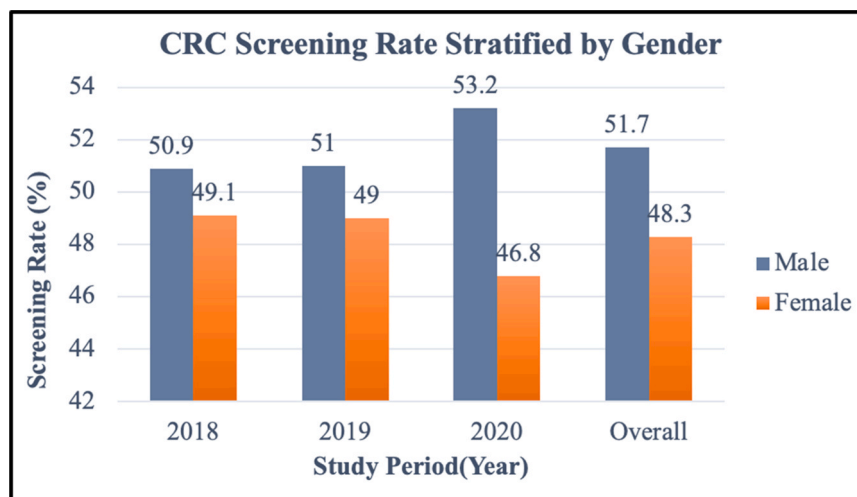
*p < 0.05,

* * < 0.01,

* * * < 0.001

OR = Odds ratio

CI = Confidence Interval

**Fig. 1.** Trend in Colorectal Cancer Screening stratified by Sex within three-year Study period. p-value: * **p < 0.05.

Our study found no significant difference in CRC screening rates between men and women over the three-year study period, indicating that both genders are equally likely to be screened. This finding aligns with a study showing similar screening rates between men and women [16]. However, it contradicts studies from Massachusetts and California, which reported higher CRC screening prevalence in men than women [23,24], and a national study indicating that men are more likely to be screened than women [25]. Our study shows a trend shift in CRC screening uptake in national data, indicating no recent gender disparity. This trend could reflect the impact of public health initiatives [55], educational efforts, and gender-inclusive screening policies [6] aimed at reducing screening barriers and promoting regular screening as a preventive measure against colorectal cancer. Hence, continued monitoring of these trends will be essential to understand if the observed trend persists over time and to identify any emerging disparities that may need to be addressed in future interventions.

Among women, race/ethnicity and education level were associated with differences in CRC screening uptake. Non-Hispanic Black women had higher screening rates than non-Hispanic White women, aligning with Shah et al.'s findings that Black women were 1.51 times more likely to undergo screening than their White counterparts [16]. Additionally, women with some college education were more likely to be screened, though this contrasts with Shah et al., who found higher screening rates among those with a bachelor's degree [16]. However, perceiving obesity as a cancer risk factor was not significantly associated with screening in the adjusted model. These findings highlight the need for tailored interventions, including culturally sensitive programs targeting women with some college education or less, to help reduce disparities and improve screening uptake.

Unique to men, our study also emphasizes the need for increased attention to specific participant characteristics, particularly sexual orientation. We found that straight men had lower odds of screening compared to gay or bisexual men, aligning with previous findings [16, 55,56]. One possible reason for this disparity is that men who have gay/bisexual men experience higher rates of anal cancer [57], which could result in greater emphasis on prioritizing colorectal cancer screening [16]. Consequently, healthcare providers may focus more on ensuring that gay/bisexual men undergo regular screening to detect potential issues at an earlier stage, thus influencing the overall approach to CRC prevention and detection efforts. Male former smokers had higher CRC screening uptake, consistent with studies showing they are more likely to undergo screening than never-smokers [25,55]. This pattern may be attributed to the heightened awareness of health risks associated with smoking, including the increased risk of colorectal cancer. As a result, male former smokers may be more motivated to engage in preventive measures such as CRC screening to detect potential issues early.

Implication for clinical practice and research

This study provides critical insights with clinical and research implications that encourage the promotion of CRC screenings. It emphasizes the need for healthcare providers to educate patients, particularly those who are overweight, about the increased cancer risk associated with obesity. Given that health insurance is a significant factor in screening, efforts could be directed towards uninsured individuals having access to screening services, possibly through policy changes or community partnerships. The study highlights the importance of race/ethnicity in screening behaviors, with non-Hispanic Black individuals, especially women, showing higher screening rates, suggesting that culturally tailored interventions could be effective in increasing participation among diverse populations. For future research, further exploration of how obesity-related cancer risk perceptions influence screening behaviors is needed, along with in-depth studies on the impact of healthcare provider recommendation and Patient-centered communication, location of the patient's residence, and accessibility to the CRC

screening and diagnosis.

Strengths and limitations

Our study utilized merged data with consistent weighting algorithms to enhance data integration. By combining multiple survey cycles, we increased the sample size, improving the study's power and precision of estimates. Our study is representative of the national demographic trends associated with obesity risk perception and screening. HINTS being a nationally representative survey; the study's findings are generalizable to the U.S. population. Furthermore, stratifying data samples for a balanced representation of minority and majority groups across sexuality ensured our research was generalizable and representative.

Our study has noteworthy limitations. The HINTS survey is cross-sectional and thus cannot infer causality. Longitudinal data and studies could offer deeper insights into this association. Individuals willing to complete the survey may differ significantly from non-responders, introducing a selection bias in the study. The study can also be biased towards certain demographics due to its reliance on landlines to gather information, thereby limiting the sample to those with landlines resulting in some data points missing leading to biases. However, the HINTS survey addresses these limitations by employing weighted procedures to better reflect the target population. Additionally, HINTS data responses are self-reported and are susceptible to recall bias. The point estimate for sexual orientation-Homosexual/Gay/Lesbian men in the adjusted model for the stratified analysis shows a strong association but has a wide confidence interval due to the small sample size, which may impact the precision of the estimate. Therefore, interpretation should be done with caution. Lastly, individuals who undergo screening may later perceive a stronger link between obesity risk perception and CRC screening behavior leading to reverse causation. However, in the future longitudinal study will be crucial in addressing these concerns.'

Conclusion

The study found that perceived obesity risk is associated with increased colon cancer screening uptake, with rates slightly surpassing the national target. Overall, age, insurance status, and race/ethnicity were key factors in screening uptake. While no gender disparity was observed in CRC screening rates, the factors influencing screening varied between genders. These insights can help guide efforts to boost screening rates through targeted, sex-specific outreach and education initiatives. Additionally, ongoing active surveillance of CRC screening recommendations can help guide efforts to reduce disparities, especially among minorities and medically vulnerable populations.

Ethics approval

The study subjects were de-identified and there was no patient contact; thus, the study is exempt from an Institutional Review Board's review.

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CRedit authorship contribution statement

Ezeigwe, Ogochukwu Juliet: Conceptualization, Methodology, Data Curation, Supervision, Formal Analysis, Writing-Review Editing. **Maria E. Fernandez:** Methodology, Supervision, Writing-Review Editing. **All other authors:** Writing-Original Draft, Writing-Review Editing

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data access requests are publicly available at <https://hints.cancer.gov/>

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